

Phishing Email Analysis and IOC Extraction Report

Project: Phishing Email Analysis and IOC Extraction Workflow

Case Study: DHL Express Shipment Notification

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Classification: Phishing Email

1. Executive Summary:

This report analyzes a phishing email that impersonates DHL Express to deceive recipients into opening a malicious attachment. "The attacker leverage brand impersonation, sender spoofing, and a malicious Excel attachment to increase credibility and deliver a malware payload."

2. Scenario Overview

The email claims to be a shipment notification from DHL Express. The message attempts to create trust by using DHL branding and shipment related language. The primary objective of the attacker is to convince the recipient to open an attached Excel document that may contain malicious macros or hidden payloads.

3. Phishing Techniques Identified

Spoofed Sender Identity:

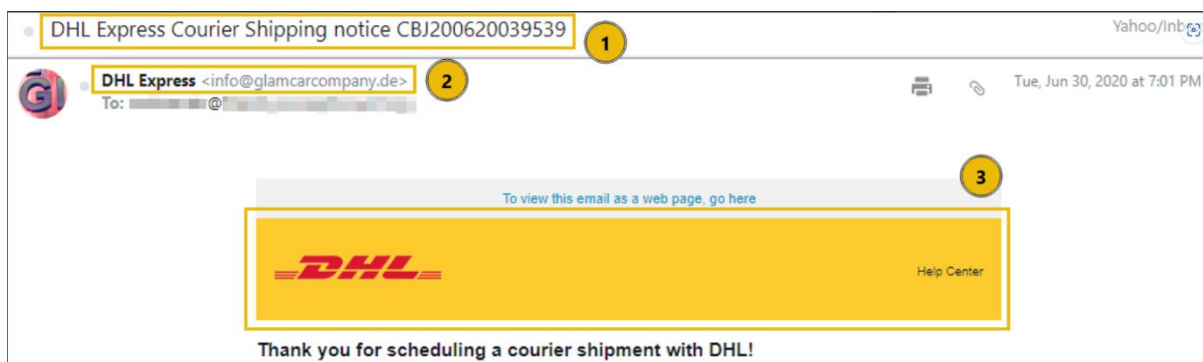
- The display name is set as "DHL Express"
- The actual sender email does not belong to the legitimate DHL domain
- The technique is used to gain the victim's trust

Brand Impersonation:

- DHL logos and HTML formatting are used
- The email visually resembles a legitimate DHL notification

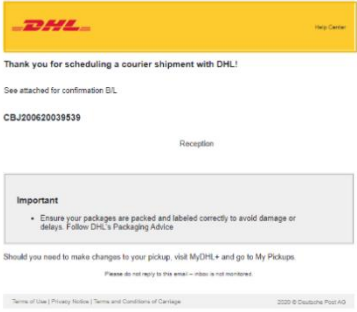
Malicious Attachment:

- The email contains an Excel attachment
- Opening the attachment could execute malicious code or download additional malware.



4. Indicators of Compromise (IOCs)

IOC Type	Value
Display Name	DHL Express
Brand Used	DHL
Attachment Type	Microsoft Excel Document
Attack Category	Phishing
Delivery Method	Email Attachment





XLSX Attachment

5. Analysis Findings

Subject Analysis:

The subject line creates urgency and encourages the recipient to review shipment details

Sender Analysis:

The display name appears legitimate; however, the underlying email address does not match DHL's official domain

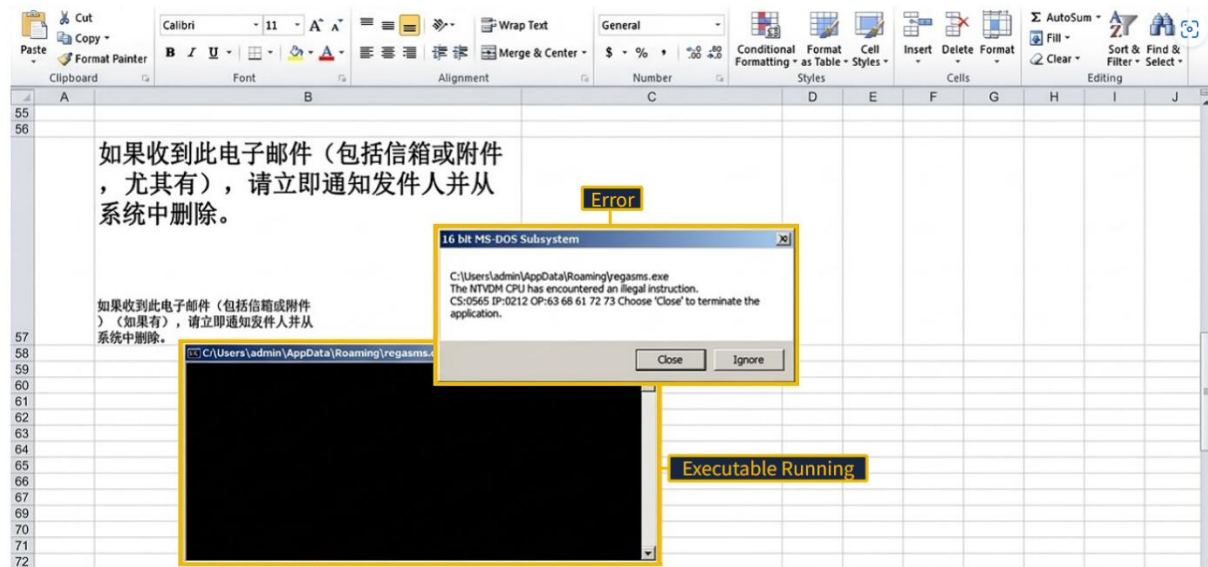
Content Analysis:

The email uses professional branding and formatting to appear authentic.

Attachment Analysis:

The attached Excel document is the primary threat vector. Such files commonly contain:

- Malicious macros
- Embedded scripts
- Malware downloaders
- Remote access trojans (RATs)



6. Detection Opportunities

- Detect suspicious attachment types.
- Scan attachments in a sandbox.
- Alert on brand impersonation.

7. Risk Assessment

Risk Level: High

Severity: High

Confidence: High

Category: Phishing/ malware Delivery

Reasons:

- Trusted brand impersonation
- Malicious attachment delivery
- Potential malware execution
- Possibility of credential theft or system compromise

8. MITRE ATT&CK Mapping

Technique ID : T1566.001

Technique Name : Phishing : Spearphishing Attachment

The attacker delivers a malicious attachment through email to persuade the victim to open the file, potentially leading to malware execution or credential compromise.

9. Recommended Actions

1. Do not open the attachment.
2. Verify shipment notification through official DHL channels.
3. Block the sender domain
4. Submit the attachment to a sandbox environment
5. Update email security controls
6. Educate users about phishing indicators

10. Conclusion

This email is a phishing attempt that abuse DHL branding to deliver a malicious Excel attachment. Multiple phishing indicators, including sender spoofing, brand impersonation, and attachment-based malware delivery, confirm that the email should be classified as malicious and blocked.